

Table 2: A breakdown of the 150 hand-evaluated examples from Natural Questions where the T5 predictions were labelled as incorrect by the automatic procedure. We found only 62% of these to be true positives.

Category	Percentage	Example		
		Question	Target(s)	T5 Prediction
True Negative	62.0%	what does the ghost of christmas present sprinkle from his torch	little warmth, warmth	confetti
Phrasing Mismatch	13.3%	who plays red on orange is new black	kate mulgrew	katherine kiernan maria mulgrew
Incomplete Annotation	13.3%	where does the us launch space shuttles from	florida	kennedy lc39b
Unanswerable	11.3%	who is the secretary of state for northern ireland	karen bradley	james brokenshire

swers do not exactly match the ground-truth context intended for each question. We therefore manually inspected 150 examples from the Natural Questions validation set where our model’s prediction was counted as incorrect in hopes of identifying “false negatives” according to the exact match metric. We found that false negatives fell into three broad categories: First, answers with meaning-preserving differences in phrasing (e.g. “April 15” vs. “April 15th”); second, questions that were missing all possible correct answers (e.g. “where does the us launch space shuttles from” was annotated with the single ground-truth answer “florida”, despite many possible correct answers such as “Kennedy Space Center”, “Merritt Island”, “Cape Canaveral”, etc.); and finally, some questions were unanswerable without knowing the exact time or article they referred to (e.g. “what is the latest version of microsoft office 2010” depends on when the question is being asked). We provide examples of each of these false negative types in table 2. We note that open-book question answering systems could also be impacted to a lesser extent by these issues (e.g. if they select a slightly different answer span from the annotated one or retrieve a non-golden document that contains a different correct answer).

Of the 150 examples inspected, we found that 20 were marked as incorrect due to differences in phrasing, another 20 were not annotated with all correct answers, and 17 were unanswerable without appropriate context. Removing unanswerable questions from the validation set and recomputing our model’s accuracy based on this false-negative rate produces a score of 57.8. This suggests that the performance of closed-book question answering systems (in terms of how often it correctly answers questions) is substantially underestimated by the evaluation procedure used in these bench-

marks. For full transparency, we publicly release the results of our human evaluation and include an appropriate reference when we determined that a predicted answer was missing from ground-truth.⁵

4 Conclusion

In this short paper, we have shown that large language models pre-trained on unstructured text can attain competitive results on open-domain question answering benchmarks without any access to external knowledge. This suggests a fundamentally different approach to designing question answering systems, motivating many threads for future work: First, we obtained state-of-the-art results only with the largest models which had around 11 billion parameters. This model size can be prohibitively expensive in resource-constrained settings, prompting future work on more efficient language models. Second, “open-book” models typically provide some indication of what information they accessed when answering a question. This can provide a useful form of interpretability. In contrast, our model distributes knowledge in its parameters in an inexplicable way and hallucinates realistic-looking answers when it is unsure. Third, the maximum-likelihood objective used to train our model provides no guarantees as to whether a model will learn a fact or not. This makes it difficult to ensure that the model obtains specific knowledge over the course of pre-training and prevents us from explicitly updating or removing knowledge from a pre-trained model. Finally, the tasks we used in this paper mainly measure “trivia”-style knowledge. We are therefore interested in measuring performance on question answering tasks that require reasoning capabilities such as DROP (Dua et al., 2019).

⁵<https://goo.gle/t5-cbqa-human-eval>

Acknowledgments

We thank Kelvin Guu, Kenton Lee, Ming-Wei Chang, Zora Tung, and Ice Pasupat for providing the open-domain question answering evaluation setup and access to their salient span-annotated data; Roy Frostig and Katherine Lee for comments and suggestions on this manuscript; Noah Constant for suggesting we try salience span masking; and Monica Dinulescu for building an interactive demonstration of our results.⁶

References

- Akari Asai, Kazuma Hashimoto, Hannaneh Hajishirzi, Richard Socher, and Caiming Xiong. 2019. Learning to retrieve reasoning paths over Wikipedia graph for question answering. *arXiv preprint arXiv:1911.10470*.
- Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. 2013. Semantic parsing on freebase from question-answer pairs. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*.
- Kurt Bollacker, Colin Evans, Praveen Paritosh, Tim Sturge, and Jamie Taylor. 2008. Freebase: a collaboratively created graph database for structuring human knowledge. In *Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data*, pages 1247–1250.
- Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading Wikipedia to answer open-domain questions. *arXiv preprint arXiv:1704.00051*.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. BoolQ: Exploring the surprising difficulty of natural yes/no questions. *arXiv preprint arXiv:1905.10044*.
- Andrew M. Dai and Quoc V. Le. 2015. Semi-supervised sequence learning. In *Advances in Neural Information Processing Systems*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Dheeru Dua, Yizhong Wang, Pradeep Dasigi, Gabriel Stanovsky, Sameer Singh, and Matt Gardner. 2019. Drop: A reading comprehension benchmark requiring discrete reasoning over paragraphs. *arXiv preprint arXiv:1903.00161*.
- Thibault F  vry, Livio Baldini Soares, Nicholas FitzGerald, Eunsol Choi, and Tom Kwiatkowski. 2020. Entities as experts: Sparse memory access with entity supervision. *arXiv preprint arXiv:2004.07202*.
- Kelvin Guu, Kenton Lee, Zora Tung, Pasupat Panupong, and Ming-Wei Chang. 2020. Realm: Retrieval-augmented language model pre-training. *arXiv preprint arXiv:2002.08909*.
- Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. *arXiv preprint arXiv:1801.06146*.
- Zhengbao Jiang, Frank F Xu, Jun Araki, and Graham Neubig. 2019. How can we know what language models know? *arXiv preprint arXiv:1911.12543*.
- Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. *arXiv preprint arXiv:1705.03551*.
- Vladimir Karpukhin, Barlas Ouguz, Sewon Min, Ledell Yu Wu, Sergey Edunov, Danqi Chen, and Wen tau Yih. 2020. Dense passage retrieval for open-domain question answering. *arXiv preprint arXiv:2004.04906*.
- Daniel Khashabi, Snigdha Chaturvedi, Michael Roth, Shyam Upadhyay, and Dan Roth. 2018. Looking beyond the surface: A challenge set for reading comprehension over multiple sentences. In *Proceedings of North American Chapter of the Association for Computational Linguistics (NAACL)*.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7.
- Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. 2019. ALBERT: A lite BERT for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942*.
- Kenton Lee, Ming-Wei Chang, and Kristina Toutanova. 2019. Latent retrieval for weakly supervised open domain question answering. *arXiv preprint arXiv:1906.00300*.
- Jeffrey Ling, Nicholas FitzGerald, Zifei Shan, Livio Baldini Soares, Thibault F  vry, David Weiss, and Tom Kwiatkowski. 2020. Learning cross-context entity representations from text. *arXiv preprint arXiv:2001.03765*.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.

⁶<http://t5-trivia.glitch.me/>

- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *EMNLP*.
- Sewon Min, Danqi Chen, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2019a. A discrete hard EM approach for weakly supervised question answering. *arXiv preprint arXiv:1909.04849*.
- Sewon Min, Danqi Chen, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2019b. Knowledge guided text retrieval and reading for open domain question answering. *arXiv preprint arXiv:1911.03868*.
- Lin Pan, Rishav Chakravarti, Anthony Ferritto, Michael Glass, Alfio Gliozzo, Salim Roukos, Radu Florian, and Avirup Sil. 2019. Frustratingly easy natural question answering. *arXiv preprint arXiv:1909.05286*.
- Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. *arXiv preprint arXiv:1802.05365*.
- Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H Miller, and Sebastian Riedel. 2019. Language models as knowledge bases? *arXiv preprint arXiv:1909.01066*.
- John Prager. 2006. Open-domain question-answering. *Foundations and Trends in Information Retrieval*, 1(2).
- Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. 2018. Improving language understanding by generative pre-training.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv preprint arXiv:1910.10683*.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. Squad: 100,000+ questions for machine comprehension of text. *arXiv preprint arXiv:1606.05250*.
- Noam Shazeer and Mitchell Stern. 2018. Adafactor: Adaptive learning rates with sublinear memory cost. *arXiv preprint arXiv:1804.04235*.
- Alon Talmor, Yanai Elazar, Yoav Goldberg, and Jonathan Berant. 2019. olympics—on what language model pre-training captures. *arXiv preprint arXiv:1912.13283*.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*.
- Guillaume Wenzek, Marie-Anne Lachaux, Alexis Conneau, Vishrav Chaudhary, Francisco Guzman, Armand Joulin, and Edouard Grave. 2019. Ccnet: Extracting high quality monolingual datasets from web crawl data. *arXiv preprint arXiv:1911.00359*.
- Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Ruslan Salakhutdinov, and Quoc V. Le. 2019. XLNet: Generalized autoregressive pretraining for language understanding. *arXiv preprint arXiv:1906.08237*.
- Sheng Zhang, Xiaodong Liu, Jingjing Liu, Jianfeng Gao, Kevin Duh, and Benjamin Van Durme. 2018. ReCoRD: Bridging the gap between human and machine commonsense reading comprehension. *arXiv preprint arXiv:1810.12885*.

A Metrics for Natural Questions

Compared to WebQuestions and TriviaQA, Natural Questions is distributed with a much richer set of annotations: Each question can be annotated either as unanswerable (given the oracle context), with a short answer, or with a yes/no answer; questions in the validation set can be annotated more than once; and some questions have multiple answers (e.g. “Who are the members of the Beatles?” has four answers). We consider two variants of Natural Questions. In both cases, we omit the “unanswerable” label and long answers, which are nearly impossible to predict without the oracle context.

The first variant is the standard “open-domain” version as used e.g. by (Lee et al., 2019; Min et al., 2019b,a; Asai et al., 2019), where 1) the model is only ever trained to output a single answer; 2) if a question has multiple answers, it is only trained to predict the first answer; 3) any questions with answers longer than five tokens are ignored; 4) answers are normalized before being compared (in the same manner as is typically done for WebQuestions and SQuAD); and 5) a predicted answer is considered correct if it matches any of the answers provided by any of the annotators (e.g. “Ringo Starr” would be considered a correct answer to “Who are the members of the Beatles?”).

The second variant closely matches the official evaluation procedure used by the Natural Questions leaderboard, where our model is trained to predict all ground-truth answers and is only considered correct if it predicts *all* answers for any one of the annotators. As in the official evaluation, we consider questions with fewer than two non-null annotations unanswerable (given the context), but because we cannot predict unanswerability without the context, we only report the recall score. Further, because our model does not have access to the oracle context, we also normalize predicted and ground-truth answers when comparing them. The use of multiple possible answers also required minor modification of our text-to-text format. In this case, we trained the model to output each answer delimited by the text “answer:” (for example, “answer: John Lennon answer: Ringo Starr answer: George Harrison answer: Paul McCartney”). We then split out each answer from the model’s predictions as a postprocessing step before evaluating it against the set of answers provided by each annotation.

B Other Things We Tried

In the course of undertaking this study, we tried various ideas that ultimately did not improve performance. We briefly discuss them here.

Continued Pre-Training on Wikipedia The T5 checkpoints we used were primarily pre-trained on C4, a large and diverse dataset of unstructured web content. We were interested to see whether we could improve performance by doing further pre-training on data that was better tailored to the tasks we considered. Since both Natural Questions and TriviaQA source their answers from Wikipedia articles, we experimented with further pre-training on text data from English Wikipedia with the same unsupervised objective (“span corruption”) as was used by T5. We found that this additional “in-domain” pre-training had virtually no effect on performance. This may be because C4 already contains many articles from Wikipedia and the T5 checkpoints were pre-trained long enough to see plenty of this content.

Pre-Training From Scratch On Wikipedia Since *all* of the answers to the questions in Natural Questions appeared in Wikipedia, we carried out an additional experiment where we pre-trained T5 from scratch only on data from Wikipedia. We pre-trained on up to 1 trillion tokens (the same amount the T5 checkpoints were pre-trained on) with the span corruption objective and measured fine-tuned performance after various amounts of pre-training. Unfortunately, this resulted in dramatically worse performance regardless of the amount of pre-training. We suspect that this is because Wikipedia is too small and results in detrimental overfitting.

Span-Corruption Pre-Training on Wikipedia Sentences with Salient Spans As described previously, we observed significant performance gains with additional pre-training using “salient span masking” (SSM) on the Wikipedia sentence dataset from Guu et al. (2020) but not when using the standard “span corruption” (SC) from Raffel et al. (2019) on longer Wikipedia articles. While SC masks random spans of the input by dropping 15% of its tokens (sampled each epoch) and replacing each consecutive span of dropped tokens with a unique sentinel, SSM specifically masks out one named entity or date in the input sentence.

We were interested in determining whether the